Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [14EC43]

FOURTH SEMESTER B.TECH UNIVERSITY EXAMINATION, JUNE – 2016

MICROPRCESSORS AND MICROCONTROLLERS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. **Answer any FIVE full questions choosing at least TWO questions from each Part.**
2. **Any missing data may be suitably assumed**

PART-A

1.	a.	Distinguish between microprocessor and microcontroller.	04
	b.	Discuss the organization of internal memory RAM of 8051 with aid of neat diagram.	07
	c.	Explain the dual functionally of post-3	03
2.	a.	Discuss the addressing mode of 8051 microcontroller. i. Immediate addressing mode ii. Direct addressing mode iii. Register addressing mode iv. register indirect addressing mode	06
	b.	Memory location starting from 8100h contains 10 bytes of mixed positive and negative nos. write on ALP to separate them and store in connective memory location stating from 30h and 40h respectively.	08
3.	a.	What is assembler directive? Explain ORG, DB and EQU directives.	06
	b.	Write an ALP to find larger number in single dimensional array of length N=5.	08
4.	a.	Discuss the three factors to be considered to generate time delay in 8051 microcontroller.	03
	b.	A door sensor is connected to the pin p1.1 and a buzzer is connected to p1.7 write a 8051 C-program to monitor the door sensor when it opens sound the buzzer. Send square wave to sound the buzzer to few hundred Hz.	06
	c.	Write an 8051 c-program to convert packed BCD 0x75 to ASCII number and display the bytes on P1 and P2 respectively.	05

PART-B

5.	a.	Explain the working of timer-0 in mode-1 configuration with GATE=0 and GATE=1 along with neat sketch.	06
	b.	Assuming XTAL=11.0592MHZ, write an ALP to generate a square wave of 50Hz frequency on Pin 2.3 using timer in most suitable mode.	05
	c.	Distinguish between timer and counter.	03

6. a. What are the programming steps to be followed to transfer the data serially? **06**

b. Write a program to transfer the average "YES" serially at 9600bps, 8-bit data, 1-stop bit. Continuously. **06**

c. Explain how to double the band rate of transceiver without changing the value in TH1 register. **02**

7. a. Discuss the interrupt table and explain the types of interrupt. **05**

b. Write a C-program that continuously gets a single bit of data from p1.7 and sends it to p1.0 in the main loop, while simultaneously. **09**

i. Creating a square wave of 200ms period on pin p2.5.

ii. Sending letter 'A' to the serial port.

8. a. Interface 8051 microcontroller to 8-bit DAC and develop C-program to generate the sine wave. **06**

b. Interface 8051 microcontroller to stepper motor and develop C-program to rotate the stepper motor. Change the direction of rotation on hardware interrupt on pin INTO (resent button connected to it) **08**